

LoRaWAN IoT Platform Communication Protocol

Version History:

Version number	Release Date	Implementation Date	Creator/Modifier	Modification reason/item
V1.0	July 30, 2025	July 30, 2025	JW	

OTAA triplet:

Dev EUI: the device address code, for example 0095272510300001;

App EUI: 6D65697368756F6B;

App Key: 0930FCFB4AB7DD793259ECD6607F95B8c

ABP Triplet:

None.

1. Data transmission process

The water meter (slave unit) uploads data; upon receiving the data, the platform sends a response containing relevant information. Details regarding the transmitted and response data are provided in the table below.

2. Format of Meter Data Uploads

Serial	Data Name	Data format	Data length
1	Data header	0x68	1 byte
2	Instrument Type	0x10 - cold water meter	1 byte
3	Control code	0x01	1 byte
4	Data length	Hexadecimal	1 byte
5	Instrument Status	Reserve	1 byte
6	Valve status	Bit 0: Set to 1 - Valve open Bit 1: Set to 1 - Valve closed Bit 7: Set to 1 - Valve fault	1 byte
7	Alarm data	Reserve	2 bytes
8	Battery voltage	Floating-point number (little-endian), unit: V	4 bytes
9	Cumulated flow (integer part)	Integer (little-endian), unit: m ³	4 bytes
10	Cumulated flow (fractional part)	Floating-point number (little-endian), unit: m ³	4 bytes

11	Remaining Flow	Floating-point number (little-endian), unit: m ³	4 bytes
12	Top-up Serial Number	Hexadecimal	1 byte
13	Payment Type	Hexadecimal	1 byte
14	Reserve		8 bytes
15	Checksum	Checksum of all bytes from the data header up to (but not including) the checksum byte.	1 byte
16	Data tail	0x16	1 byte

Example: Water Meter Sending a message

6810011E000100002B9B6F4022000000D773183FD1CCB4C1010100000000000000E416

```

68          // Data header
10          // Instrument Type
01          // Control code
1E          // Data length
00          // Instrument Status
01          // Valve open
00 00      // Alarm Information
2B 9B 6F 40 // Battery voltage: 3.74 V (rounded to two decimal places)
22 00 00 00 // The integer part of the cumulative flow is 34 m³.
D7 73 18 3F // The fractional part of the cumulative flow is 0.596 m³ (rounded to three
              decimal places).
D1 CC B4 C1 // Remaining flow: -22.600 m³ (rounded to 3 decimal places)
01          // Top-up Serial Number 1
01          // Work Type: Pre-paid
00 00 00 00 00 00 00 00 00 00 // Reserve
E4          // Checksum
16          // Data tail

```

3. Platform response data frame format

Platform response data frame format (hex)

Name	Code	Data length
Start-of-frame code	0x68 (Fixed value)	1 byte

Instrument Type	0x70 (Fixed value)	1 byte
Address code	0xA0、0xA1、0xA2、0xA3、0xA4、0xA5、0xA6、0xA7 This data frame represents the water meter number.	8 bytes
Control code	Bit 0: Set to 1 - Include valve operation; Bit 1: Set to 1 - Include upload frequency modification; Bit 3: Set to 1 - Include top-up water volume; Bit 4: Set to 1 - Include operating mode modification; Bit 5: Set to 1 - Include remote reset (production testing);	2 bytes
Data length field	Calculate as required (total number of bytes from the start of the data field length up to, but not including, the checksum; excluding the data field length and checksum bytes).	1 byte
Data field		N bytes
Checksum	Checksum of all bytes from the data header up to (but not including) the checksum byte.	1 byte
Termination character	0x16	1 byte

When the control code byte is 0, there is no additional content; the response contains only time synchronization information.

When a specific bit in the control code is set to 1, the corresponding content is appended after the time synchronization information, in accordance with the definition for that bit.

The additional content is arranged according to the bit order; if a preceding bit is set to 0, the subsequent additional content shifts forward to fill the gap.

The format for the corresponding content is defined as follows:

Serial	Operation Code	Operational Details
1	All bits set to 0 - Time synchronization	6-byte BCD code; year-month-day-hour-minute-second

2	Bit 0: Set to 1 - Valve operation	0x00: Close valve; 0x01: Open valve; 1 byte
3	Bit 1: Set to 1 - Modify upload frequency	Hexadecimal data, in minutes, 2 bytes, little-endian format. The minimum interval is 5 minutes.
4	Bit 3: Set to 1 - Top-up water volume	5 bytes: top-up sequence number (1 byte) + top-up water volume (floating-point, 4 bytes, little-endian).
5	Bit 4: Set to 1 - Modify operating mode	0x00: Postpaid; 0x01: Prepaid. 1 byte.
6	Bit 5: Set to 1 - Remote reset	See the detailed explanation below.

3.1 Examples of time synchronization commands

Calibrating the water meter time to 26-08-25 10:58:12.

Platform-issued / Distributed by the platform:

68700095272510300001000006260825105812CD16

68 70 // Fixed value

00 95 27 25 10 30 00 01 // Address code

00 00 // Operation Code

06 // Data field length

26 08 25 10 58 12 //Time 26-08-25 10:58:12

CD // Checksum

16 // Data tail

3.2 Examples of valve open/close commands

1	Bit 0: Set to 1 - Valve operation	0x00: Close valve; 0x01: Open valve. 1 byte.
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The platform issues a valve-opening command.:

6870009527251030000100010726082510581201D016

68 70 // Fixed value

00 95 27 25 10 30 00 01 // Address code

00 01 // Operation Code

07 // Data field length

26 08 25 10 58 12 //Time

01 // Open the valve

D0 // Checksum
16 // Data tail

3.3 Examples of commands for modifying upload frequency

1	Bit 1: Set to 1 - Modify upload frequency	Hexadecimal data, in minutes, 2 bytes, little-endian format, minimum interval of 5 minutes.
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The platform issues updates at a frequency of 10 minutes. :

687000952725103000010002082608251058120A00DB16

68 70 // Fixed value

00 95 27 25 10 30 00 01 // Address code

00 02 // Operation Code

08 // Data field length

26 08 25 10 58 12 // Time

0A 00 // The modification frequency is 8 hours.

DB // Checksum

16 // Data tail

3.4 Examples of water credit top-up commands

1	Bit 3: Set to 1 - Top-up water volume	5 bytes: top-up sequence number (1 byte) + top-up water volume (floating-point, 4 bytes, little-endian).
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The platform issued a top-up water volume of 10.0 m³:

6870009527251030000100080B26082510581201000020413C16 // top up 10.0 m³

68 70 // Fixed value

00 95 27 25 10 30 00 01 // Address code

00 08 // Operation Code

0B // Data field length

26 08 25 10 58 12 // Time

01 //Top up serial 01

00002041 //10.0 m³

3C // Checksum

16 // Data tail

3.5 Examples of commands for changing the operating mode

1	Bit 4: Set to 1 - Modify operating mode	0x00: Postpaid; 0x01: Prepaid. 1 byte.
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The platform employs a post-paid operational model.:

6870009527251030000100100726082510581200DE16

68 70 // Fixed value

00 95 27 25 10 30 00 01 // Address code

00 10 // Operation Code

07 // Data field length

26 08 25 10 58 12 // Time

00 // Change to postpaid

DE // Checksum

16 // Data tail

3.6 Example of command to reset cumulative water volume (for production testing only)

1	Bit 5: Set to 1 - Remote clear (production test)	See the detailed explanation below.
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Platform-issued: 6870009527251030000100200A26082510581200000000F116

4. Water Meter Response Control Data Frame Format

serial	Data Name	Data format	Data length
1	Data header	0x68	2 Bytes
2	Instrument Type	0x10 - 冷水表	1Byte
3	Control code	0x02	1Byte
4	Data field length	16 (0x10)	1Byte
5	Data field	Opcode sent by the server	2 Bytes
		Cumulative flow (integer part) (same as above)	4 Bytes
		Cumulative flow (fractional part) (same as above)	4 Bytes
		Remaining data allowance (same as above)	4 Bytes
		Top-up Serial Number	1Byte
		Payment type	1Byte

		Reserve	3 Bytes
6	Checksum	Checksum of all bytes from the control code up to (but not including) the checksum byte.	1 Byte
7	Data tail	0x16	1Byte

The following is an example of the data returned by the water meter after receiving a valve control command from the platform. :

68100210000122000000D773183FD1CCB4C10101EA16

```

68          // Data header
10          // Instrument type
02          // Control code
10          // Data field length
00 01      // Valve control opcode sent by the server
22 00 00 00 // Cumulated Flow (Integer Part) 34 m³
D7 73 18 3F // Cumulated Flow (Fractional Part) 0.596 m³
D1 CC B4 C1 // Remaining FLOW -22.600 m³
01          // Top up serial 01
01          // Work Type: Pre-paid
EA          // Checksum
16          // Data tail

```